		Precision	Recall	F1-score
SVM + HOG	Accuracy			0.60
	Macro avg	0.49	0.43	0.44
	Weighted avg	0.60	0.60	0.59
ResNET Baseline 128*128, Adam	Accuracy			0.73
	Macro avg	0.71	0.60	0.60
	Weighted avg	0.78	0.73	0.72
ResNET Ablation1 128*128, SGD	Accuracy			0.94
	Macro avg	0.93	0.91	0.92
	Weighted avg	0.95	0.94	0.94
ResNET Ablation 2 64*64, Adam	Accuracy			0.71
	Macro avg	0.66	0.59	0.60
	Weighted avg	0.74	0.71	0.71
ViT 224*224, Adam	Accuracy			0.90
	Macro avg	0.89	0.88	0.87
	Weighted avg	0.92	0.90	0.90

Table 1. Quantitative performance comparison of various classification methods on the Caltech-101 dataset.

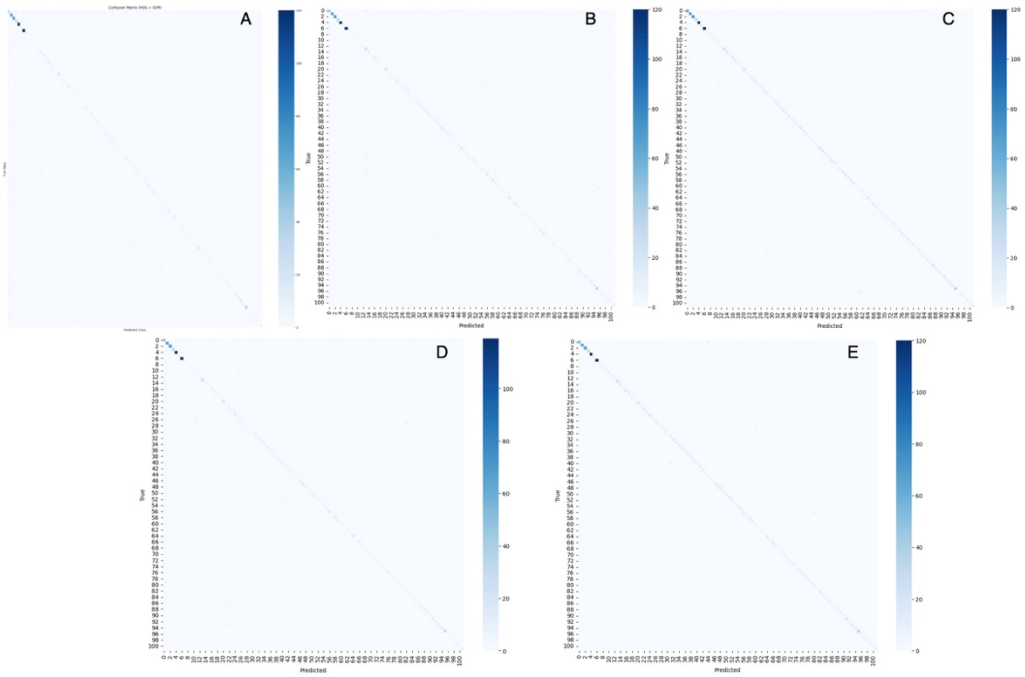

Fig 1. Confusion matrices visualizing the classification results across five different experimental setups:

(A) Classical HOG + SVM. (B) ResNET baseline - 128*128, Adam. (C) ResNET ablation 1 - 128*128, SGD. (D) ResNET ablation 2 - 64*64, Adam. (E) Vision Transformer - 224*224, Adam.

Result

The experimental results demonstrate that ResNet Ablation 1 (128x128, SGD) achieved the highest performance, reaching 0.94 Accuracy and a 0.92 Macro F1-score. This reflects a significant 21% improvement over the Adam-based ResNet baseline, which recorded 0.73 Accuracy. The classical HOG+SVM method produced the lowest results with 0.60 Accuracy.

In the ablation studies, reducing the image resolution to 64x64 (Ablation 2) caused accuracy to drop from 0.73 to 0.71, suggesting that fine-grained spatial features are necessary for distinguishing between the 101 categories. While the Vision Transformer (ViT) performed well at 0.90 Accuracy, it did not surpass the optimized ResNet model. These performance gaps are visually represented in the confusion matrices in Figure 1, where the best-performing model (C) displays the most distinct diagonal alignment.

Conclusion

This project highlights the clear superiority of deep learning over classical feature extraction methods like HOG+SVM for complex image classification tasks. A key takeaway is that hyperparameter optimization (specifically the choice of the SGD optimizer) had a more substantial impact on final accuracy than adopting a newer architecture like ViT. Furthermore, maintainting a higher input resolution proved vital for capturing the intricate details of the Caltech-101 dataset. Future work should focus on addressing class imbalance to improve the Macro F1-scores of underrepresented labels.

Github link